Title

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Abstract

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1 Introduction

1.1 Preliminaries

This subsection collects the notions on graphs and non-negative matrices used for both models, following the course slides [2, 3, 1].

1.1.1 Graphs

A *weighted digraph* $G = (V, E, \{a_{ij}\}_{(i,j)\in E})$ has node set $V = \{1, 2, \ldots, n\}$, edge set $E \subseteq V \times V$ and a weight $a_{ij} > 0$ for each edge $(i, j) \in E$; an edge (i, i) is a *self-loop*. Throughout the report the nodes are agents, and an edge (i, j) means that agent i is influenced by agent j. This is the convention of the course slides [1]; the tutorial by Proskurnikov and Tempo draws the edges in the opposite direction [5].

Definition 1.1 (Weighted adjacency matrix [2])**.** Given a weighted digraph G of order n, its *weighted adjacency matrix* is the $n \times n$ matrix A whose (i, j) entry is equal to the edge weight a_{ij} if $(i, j) \in E$, and is equal to zero otherwise.

Definition 1.2 (Strong connectivity and globally reachable nodes [2])**.** A digraph $G = (V, E)$ is *strongly connected* if there is a directed walk from any node to any other node. A node of G is *globally reachable* if it can be reached from any other node through a directed walk.

With our convention, a directed walk from i to j means that agent i is influenced by agent j, either directly or through intermediaries. Hence G is strongly connected when every agent influences every other agent, at least indirectly, and a globally reachable node is an agent that influences the whole group.

Definition 1.3 (Periodic digraph [2])**.** A strongly connected digraph is *periodic* if there exists an integer $k > 1$ (the period) that divides the length of every simple cycle of the digraph; otherwise it is *aperiodic*.

Equivalently, a strongly connected digraph G is periodic if and only if the greatest common divisor of the lengths of its simple cycles is larger than 1 [2]. In particular, a strongly connected digraph with a self-loop is aperiodic, since the self-loop is a cycle of length 1.

Definition 1.4 (Strongly connected components and condensation digraph [1])**.** Given a weighted digraph G, a subgraph S is a *strongly connected component* of G if S is strongly connected and any other subgraph of G strictly containing S is not strongly connected. The *condensation digraph* $C(G)$ is obtained by collapsing each strongly connected component of G into a single node: for any two distinct strongly connected components S_k and S_h, there is an edge (S_k, S_h) in $C(G)$ if and only if there exists an edge $(i, j) \in E$ with $i \in S_k$ and $j \in S_h$ (self-loops are not allowed in $C(G)$). A node of $C(G)$ with no outgoing edges is called a *sink* [2].

A sink of $C(G)$ is a group of agents that are influenced only by members of the group itself: no agent outside the group can change their opinions.

If G contains a globally reachable node, then $C(G)$ has a unique sink [1].

1.1.2 Matrices

Definition 1.5 (Row-stochastic matrix [3])**.** A matrix $A \in \mathbb{R}^{n\times n}$ is *row-stochastic* if it is non-negative and $A\mathbf{1}_n = \mathbf{1}_n$, i.e. each row sums to 1.

Theorem 1.6 (Spectral properties of row-stochastic matrices [3])**.** *If $A \in \mathbb{R}^{n\times n}$ is row-stochastic, then $1 \in \operatorname{spec}(A)$ and $\rho(A) = 1$; hence $\operatorname{spec}(A)$ is contained in the closed unit disk.*

Definition 1.7 (Semi-convergent and convergent matrices [3])**.** A matrix $A \in \mathbb{R}^{n\times n}$ is *semi-convergent* if $\lim_{k\to+\infty} A^k$ exists and is finite, and *convergent* (or *Schur stable*) if, in addition, $\lim_{k\to+\infty} A^k = 0$.

Theorem 1.8 (Convergence and spectral radius [3])**.** *A matrix $A \in \mathbb{R}^{n\times n}$ is convergent if and only if $\rho(A) < 1$.*

The properties of a non-negative matrix that determine its spectrum are characterized by the graph notions above.

Definition 1.9 (Irreducible and primitive matrices [3])**.** A non-negative matrix $A \in \mathbb{R}^{n\times n}$, $n \geq 2$, is *irreducible* if $\sum_{k=0}^{n-1} A^k$ is positive, and *primitive* if there exists $k \in \mathbb{N}$ such that A^k is positive.

Theorem 1.10 (Irreducible matrices and digraphs [2])**.** *Let G be a weighted digraph of order $n \geq 2$ with weighted adjacency matrix A. Then A is irreducible if and only if G is strongly connected.*

Theorem 1.11 (Primitive matrices and digraphs [2])**.** *Let G be a weighted digraph of order $n \geq 2$ with weighted adjacency matrix A. Then A is primitive if and only if G is strongly connected and aperiodic. In particular, if G is strongly connected and has at least one self-loop, then A is primitive.*

Given a weighted digraph G with weighted adjacency matrix A, the *weighted out-degree* of node i is the i-th row sum $\sum_{j=1}^{n} a_{ij}$ of A [2]. A row-stochastic matrix has all weighted out-degrees equal to 1; when some of them are smaller than 1, the following notions apply.

Definition 1.12 (Row-substochastic matrix [1])**.** A non-negative matrix $A \in \mathbb{R}^{n\times n}$ is *row-substochastic* if all its row sums are at most 1 and at least one row sum is strictly less than 1.

Lemma 1.13 (Convergent row-substochastic matrices [1])**.** *Let G be a weighted digraph with row-substochastic adjacency matrix A. Then A is convergent if and only if G contains directed walks from each node with (weighted) out-degree 1 to a node with (weighted) out-degree strictly less than 1.*

2 The French–Harary–DeGroot model

2.1 Introduction to the FDG model

2.1.1 The FDG model

Consider a group of individuals $V = \{1, 2, \ldots, n\}$, where $x_i \in \mathbb{R}$ is the opinion that individual i holds on a given topic. The influences among the individuals are described by a weighted digraph $G = (V, E, \{a_{ij}\}_{(i,j)\in E})$: following the convention of Section 1.1, an edge $(i, j) \in E$, with weight $a_{ij} > 0$, means that individual i is influenced by individual j.

The French–Harary–DeGroot (FDG) model [1], called French–DeGroot model in [5], describes how each individual, or agent, updates its opinion: the new opinion is a weighted average of the opinions of the agents it is influenced by,

$$x_i(k+1) = \sum_{j=1}^{n} a_{ij}\, x_j(k), \qquad \sum_{j=1}^{n} a_{ij} = 1, \tag{1}$$

or, in vector form, $x(k+1) = Ax(k)$, where A is the weighted adjacency matrix of G (Definition 1.1). The constraint on the weights means that A is row-stochastic (Definition 1.5).

Since $x(k) = A^k x(0)$, the asymptotic behaviour of the model is determined by the powers of A. Three outcomes are possible:

- *no convergence*: $\lim_{k\to+\infty} x(k)$ does not exist for some $x(0)$; this happens when A is not semi-convergent (Definition 1.7), e.g. when A has an eigenvalue $\mu \neq 1$ with $|\mu| = 1$;
- *consensus*: $\lim_{k\to+\infty} x(k) = \alpha \mathbf{1}_n$ for some $\alpha \in \mathbb{R}$;
- *disagreement*: $\lim_{k\to+\infty} x(k) = v$ with $v \neq \alpha \mathbf{1}_n$ for every $\alpha \in \mathbb{R}$.

2.1.2 Consensus and social power

The following theorem characterizes when the FDG model reaches a consensus. Its condition (A3) involves the subgraph of globally reachable nodes. This set of nodes is always strongly connected [1], since a walk between two globally reachable nodes only visits nodes that are themselves globally reachable. Hence its periodicity is well defined (Definition 1.3).

Theorem 2.1 (Consensus in DT averaging systems [1])**.** *Let G be a weighted digraph with row-stochastic adjacency matrix A. The following statements are equivalent:*

(A1) the eigenvalue 1 is simple and all other eigenvalues μ satisfy $|\mu| < 1$;

(A2) A is semi-convergent and $\lim_{k\to+\infty} A^k = \mathbf{1}_n w^\top$, where $w \in \mathbb{R}^n$ satisfies $w \geq 0$, $\mathbf{1}_n^\top w = 1$ and $w^\top A = w^\top$;

(A3) G contains a globally reachable node and the subgraph of globally reachable nodes is aperiodic.

If any, and therefore all, of the previous conditions are satisfied, then

(i) w is the left eigenvector of A associated with the eigenvalue 1, and $w_i > 0$ if and only if node i is globally reachable;

(ii) the solution of the FDG model satisfies $\lim_{k\to+\infty} x(k) = \left(w^\top x(0)\right) \mathbf{1}_n$.

Corollary 2.2 (Average consensus [1])**.** *If A is doubly-stochastic, i.e. row-stochastic with also $\mathbf{1}_n^\top A = \mathbf{1}_n^\top$, and (A3) holds, then $w = \frac{1}{n}\mathbf{1}_n$ and $\lim_{k\to+\infty} x(k) = \left(\frac{1}{n}\sum_{i=1}^n x_i(0)\right) \mathbf{1}_n$.*

The entry w_i is called the *social power* of agent i: it is the weight of the initial opinion $x_i(0)$ in the final opinion of the group [5].

Point (i) has a direct social interpretation. Since the consensus value $w^\top x(0)$ is the same for every agent, the initial opinion of agent i can contribute to it only if it reaches *all* the agents. If i is not globally reachable, there is at least one agent that is influenced by i neither directly nor through intermediaries: the opinion of i never reaches that agent, so it cannot enter the common value, and $w_i = 0$. Conversely, the opinion of a globally reachable agent reaches the whole group and has a positive weight. Hence the consensus value depends only on the initial opinions of the globally reachable agents.

Among the globally reachable agents, the social power depends on their position in the network, and it can be regarded as a centrality measure [5]. For instance, in the model of French with $n = 3$ agents and

$$A = \begin{bmatrix} 1/2 & 1/2 & 0 \\ 1/3 & 1/3 & 1/3 \\ 0 & 1/2 & 1/2 \end{bmatrix},$$

agents 1 and 3 listen to themselves and to agent 2, while agent 2 listens to everybody. All agents are globally reachable, but agent 2, the only one listened to by all the others, has the largest social power: $w = (2/7,\, 3/7,\, 2/7)^\top$ [5].

2.1.3 Disagreement

Theorem 2.1 requires a single globally reachable group, i.e. a unique sink in the condensation digraph $C(G)$. We now consider the case in which $C(G)$ has $n_s \geq 2$ sinks. Each sink is a strongly connected component, so its periodicity is well defined (Definition 1.3).

Theorem 2.3 (Disagreement in DT averaging systems [1])**.** *Let G be a weighted digraph with row-stochastic adjacency matrix A. Assume that the condensation digraph $C(G)$ has $n_s \geq 2$ sinks. The following statements are equivalent:*

(A1) the eigenvalue 1 is semi-simple with multiplicity n_s and all other eigenvalues μ satisfy $|\mu| < 1$;

(A2) A is semi-convergent;

(A3) each sink of $C(G)$ is aperiodic.

If any, and therefore all, of the previous conditions are satisfied, then the solution of the FDG model satisfies $\lim_{k\to+\infty} x(k) = v \neq \alpha \mathbf{1}_n$ for every $\alpha \in \mathbb{R}$.

The final opinions can be described group by group. The agents of a sink are not influenced by anyone outside the sink, so each sink evolves as an FDG model on its own and, being aperiodic, reaches an internal consensus that depends only on the initial opinions of its members. Every agent outside the sinks converges instead to a convex combination of the consensus values of the sinks [1].

In social terms, a sink is a closed group that listens only to itself, and two closed groups never influence each other: each of them keeps its own consensus value. The agents in between listen, directly or through intermediaries, to several closed groups and settle on an intermediate opinion between their consensus values. As a consequence, the disagreement is bounded: every final opinion lies between the smallest and the largest consensus value of the sinks.

2.1.4 Rosetta stone

The results of this section, together with the consensus and disagreement theorems above, classify all the possible behaviours of the FDG model. The classification is summarized in Table 1, the so-called *Rosetta stone* [1], which translates each property of the row-stochastic matrix A into a property of the digraph G and into the corresponding asymptotic outcome; for each case, the table also reports an example digraph, its matrix and its spectrum.

The two consensus cases illustrate Theorem 2.1. In case (i) both agents are globally reachable and A is doubly-stochastic, so by Corollary 2.2 $w = (1/2, 1/2)^\top$ and the group agrees on the average $\big(x_1(0) + x_2(0)\big)/2$. In case (iii) agent 2 is influenced by agent 1 but not vice versa, so only agent 1 is globally reachable: $w = (1, 0)^\top$ and the group agrees on $x_1(0)$, whatever the initial opinion of agent 2.

Case (iv) illustrates Theorem 2.3 in its simplest form: each agent is influenced only by itself, so each agent is a sink of $C(G)$ ($n_s = 2$), both sinks are aperiodic thanks to the self-loops, and every agent keeps its initial opinion, $x(k) = x(0)$ for all k.

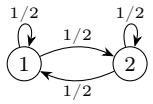
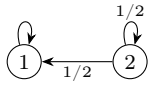
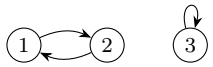
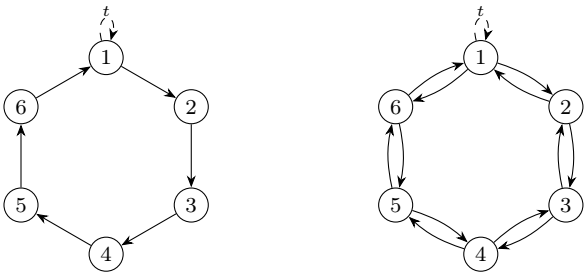
Case	Example digraph	A	spec(A)	Property of A / of G	Outcome
(i)		$\begin{bmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{bmatrix}$	$\{1, 0\}$	A primitive / G strongly connected and aperiodic	consensus
(ii)		$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$	$\{1, -1\}$	A irreducible, not primitive / G strongly connected and periodic	no convergence
(iii)		$\begin{bmatrix} 1 & 0 \\ 1/2 & 1/2 \end{bmatrix}$	$\{1, 1/2\}$	1 simple and strictly dominant, A not primitive / $C(G)$ has a unique sink, which is aperiodic	consensus
(iv)		$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$	$\{1, 1\}$	1 semi-simple but not simple, strictly dominant / $C(G)$ has at least two sinks, all aperiodic	disagreement
(v)		$\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$\{1, 1, -1\}$	1 not strictly dominant / G not strongly connected, at least one sink of $C(G)$ is periodic	no convergence

Table 1: Rosetta stone of the FDG model $x(k+1) = Ax(k)$, with A row-stochastic: for each case, an example digraph (unlabelled edges have weight 1), its matrix A, the spectrum of A, the general property of A and of G, and the outcome. Cases (i)–(iv) and the examples follow [1]; the properties of case (v), left blank in the slides, follow from [5].

2.2 Stubbornness and social power in the FDG model

In the FDG model the simplest way to account for stubbornness is the self-weight a_{ii}, i.e. the weight of the self-loop of agent i. This section studies how self-weights, and stubborn agents in particular, affect the two aspects introduced in Section 2.1: whether the group reaches an agreement, and how the social power is distributed among the agents.

All the examples use the two networks of Figure 1, both with $n = 6$ agents. In the *directed cycle* each agent is influenced only by the next one, $a_{i,i+1} = 1$; in the *bidirectional ring* each agent is influenced by its two neighbours with equal weights, $a_{i,i-1} = a_{i,i+1} = 1/2$ (indices modulo 6). Socially, in the first network everyone listens to a single person, while in the second everyone listens to the two people next to them. Without self-loops both digraphs are strongly connected and periodic: the directed cycle has a single cycle, of length 6, while every cycle of the bidirectional ring has even length, so its period is 2. Hence neither network reaches an agreement (case (ii) of Table 1). Moreover, both matrices are doubly-stochastic, so $w = \frac{1}{6}\mathbf{1}_6$ is the left eigenvector associated with the eigenvalue 1: no agent is favoured by the structure of the network, and any asymmetry in the social power will be due only to the self-weight added to agent 1.



(a) directed cycle (b) bidirectional ring

Figure 1: The two networks used in Sections 2.2 and 3. An edge $i \to j$ means that agent i is influenced by agent j. In (a) all edges have weight 1, in (b) all edges have weight 1/2. The dashed self-loop of agent 1, of weight t, is the only modification considered: the other edges of agent 1 are scaled by $1 - t$ (equation (2)).

2.2.1 Modelling stubbornness: self-weights and stubborn agents

The self-weight a_{ii} is the weight that agent i gives to its own *current* opinion [4]; it indicates how open the agent is to the opinions of the others [5]. An agent with $a_{ii} = 0$ is open-minded and relies completely on the others; an agent with a positive self-weight changes its opinion more slowly, but it still moves towards the opinions it listens to. In social terms, a self-weight is a form of inertia, not an anchor to a fixed opinion.

Definition 2.4 (Stubbornness [5])**.** An agent is *stubborn* if its opinion remains unchanged independently of the others' opinions. In the FDG model, agent i is stubborn, i.e. $x_i(k) = x_i(0)$ for all k, if and only if $a_{ii} = 1$ (and hence $a_{ij} = 0$ for all $j \neq i$).

Stubbornness is therefore the extreme case of the self-weight, and there is no intermediate degree of it: an agent with $0 < a_{ii} < 1$ is not stubborn, it only has a positive self-weight.

In the digraph, a stubborn agent has no edges towards the other agents: it is influenced by nobody, while the agents that listen to it keep doing so. Hence it forms a strongly connected component made of a single node, which is a sink of $C(G)$ and is aperiodic because of its self-loop. If, in addition, every agent is influenced by the stubborn agent i, directly or through intermediaries, then i is the only globally reachable node, and Theorem 2.1 gives $w = e_i$: the whole group converges to the initial opinion of the stubborn agent (see also [5]).

To move continuously from no self-weight to stubbornness, we modify only the row of agent 1. Given a row-stochastic matrix A with $a_{11} = 0$, as in both networks of Figure 1, we replace its first row by

$$a_1(t)^\top = (1 - t)\, a_1^\top + t\, e_1^\top, \qquad t \in [0, 1], \tag{2}$$

where e_1 is the first vector of the canonical basis, and leave the other rows unchanged. Agent 1 gives weight t to itself and splits the remaining weight $1 - t$ among the agents it listened to before, in the same proportions. The parameter t is thus the self-weight a_{11}: $t = 0$ is the original network, $0 < t < 1$ gives agent 1 a positive self-weight, and $t = 1$ makes agent 1 stubborn. For the directed cycle the first row becomes $(t,\, 1-t,\, 0,\, 0,\, 0,\, 0)$; for the bidirectional ring it becomes $(t,\, \frac{1-t}{2},\, 0,\, 0,\, 0,\, \frac{1-t}{2})$. In the following we consider the three values $t = 0$, $t = 1/2$ and $t = 1$.

2.2.2 Self-weights, stubbornness and convergence

The three values of t correspond to three different cases of the Rosetta stone.

- $t = 0$: both digraphs are strongly connected and periodic, so neither network converges (case (ii)).

- $0 < t < 1$: agent 1 still listens to its neighbours with total weight $1 - t > 0$, so the digraph stays strongly connected, and the self-loop of agent 1 is a cycle of length 1, so the digraph becomes aperiodic. By Theorem 1.11 the matrix is primitive, and the group reaches a consensus (case (i)).
- $t = 1$: agent 1 is stubborn. In both networks every other agent is influenced by agent 1 through the cycle, directly or through intermediaries, so by the discussion in Section 2.2.1 agent 1 is the only sink of $C(G)$ and the only globally reachable node, so the group reaches a consensus on its initial opinion $x_1(0)$ (case (iii)).

The same picture can be read on the spectrum of the matrix. For the directed cycle the characteristic polynomial is

$$p(\lambda) = (\lambda - 1)\left[\lambda^5 + (1-t)(\lambda^4 + \lambda^3 + \lambda^2 + \lambda + 1)\right].$$

Apart from $\lambda_1 = 1$, the eigenvalues are the roots of a polynomial of degree 5, which for a generic t have no closed-form expression; for the three values of t they are

- $t = 0$: $\lambda_1 = 1$, $\lambda_{2,3} = \frac{1}{2} \pm \frac{\sqrt{3}}{2}\, i$, $\lambda_{4,5} = -\frac{1}{2} \pm \frac{\sqrt{3}}{2}\, i$, $\lambda_6 = -1$ (the sixth roots of unity, all on the unit circle);
- $t = 1/2$: $\lambda_1 = 1$, $\lambda_{2,3} \approx 0.535 \pm 0.751\, i$, $\lambda_{4,5} \approx -0.373 \pm 0.759\, i$, $\lambda_6 \approx -0.823$;
- $t = 1$: $\lambda_1 = 1$, $\lambda_2 = \cdots = \lambda_6 = 0$.

For the bidirectional ring the characteristic polynomial is

$$p(\lambda) = (\lambda - 1)\big(\lambda - \tfrac{1}{2}\big)\big(\lambda + \tfrac{1}{2}\big)\Big[\lambda^3 + (1-t)\lambda^2 - \tfrac{2t+1}{4}\,\lambda + \tfrac{t-1}{4}\Big].$$

The eigenvalues 1 and $\pm\frac{1}{2}$ do not depend on t, while the other three are the roots of the cubic factor, whose closed-form expression in t is too involved to be informative; for the three values of t the eigenvalues are

- $t = 0$: $\lambda_1 = 1$, $\lambda_{2,3} = \frac{1}{2}$, $\lambda_{4,5} = -\frac{1}{2}$, $\lambda_6 = -1$ (on the unit circle);
- $t = 1/2$: $\lambda_1 = 1$, $\lambda_2 \approx 0.623$, $\lambda_3 = \frac{1}{2}$, $\lambda_4 \approx -0.223$, $\lambda_5 = -\frac{1}{2}$, $\lambda_6 \approx -0.901$;
- $t = 1$: $\lambda_1 = 1$, $\lambda_2 = \frac{\sqrt{3}}{2}$, $\lambda_3 = \frac{1}{2}$, $\lambda_4 = 0$, $\lambda_5 = -\frac{1}{2}$, $\lambda_6 = -\frac{\sqrt{3}}{2}$.

In both networks, for $t = 0$ some eigenvalues different from 1 lie on the unit circle, and they cause the oscillations; for $t > 0$ all the eigenvalues different from 1 lie strictly inside the unit disk, and the group converges. For $t = 1$ the directed cycle is left with the eigenvalue 0 only, which is why its convergence is reached in a finite number of steps.

Figure 2 shows the opinions $x_i(k)$ in the six cases. For $t = 0$ the opinions keep rotating along the directed cycle and keep oscillating between two configurations on the bidirectional ring. For $t = 1/2$ both networks reach a consensus whose value is a weighted average of all the initial opinions, with weights given by the social power (Section 2.2.3). For $t = 1$ both networks reach a consensus on $x_1(0)$, but in different ways. In the directed cycle every agent other than agent 1 simply copies the opinion of the next agent, so the opinion of agent 1 travels backwards along the chain, one agent per step, and the consensus is reached exactly after $n - 1 = 5$ steps. In the bidirectional ring every agent averages the opinions of its two neighbours, so the opinions approach $x_1(0)$ gradually, the neighbours of agent 1 first.

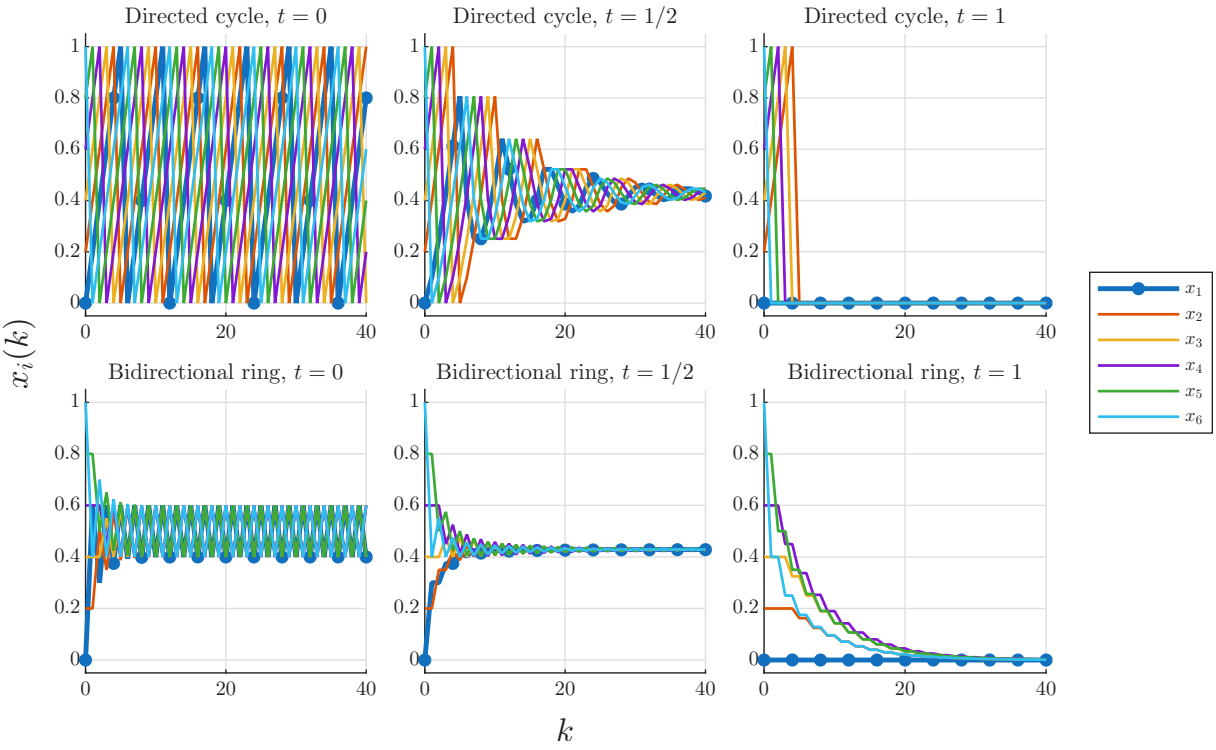


Figure 2: Opinions $x_i(k)$ in the two networks of Figure 1 (top: directed cycle; bottom: bidirectional ring) for three values of the self-weight t of agent 1 (columns), with $x(0) = (0, 0.2, 0.4, 0.6, 0.8, 1)^\top$. Agent 1 is marked with dots.

In social terms, a group in which nobody gives weight to its own opinion can keep passing opinions around forever. It is enough that a single agent gives some weight to its current opinion, without being stubborn, for the whole group to reach an agreement. If that agent becomes stubborn, the agreement is reached on its own opinion. The structure of the network decides only how this happens: by copying along a chain in the directed cycle, by averaging with the neighbours in the bidirectional ring.

2.2.3 Self-weights, stubbornness and social power

For $t > 0$ both networks reach a consensus (Section 2.2.2), so the social power w of Section 2.1.2 is well defined, and Theorem 2.1 describes how it depends on t.

- $0 < t < 1$: the digraph is strongly connected, so every agent is globally reachable and, by point (i) of Theorem 2.1, every agent has a positive social power. Agent 1 gives weight to its own opinion, but it does not take all the influence: the initial opinion of every agent enters the consensus value.

- $t = 1$: agent 1 is stubborn and is the only globally reachable node, so $w = e_1$: the stubborn agent takes all the social power, and the initial opinions of the other agents have no effect on the final opinion of the group.

Between these two extremes, the social power can be computed as the left eigenvector of $A(t)$ associated with the eigenvalue 1, normalized so that its entries sum to 1 (condition (A2) of Theorem 2.1). Figure 3 reports the result for $t \in (0, 1]$. Numerically, the social power of agent 1 grows with t towards 1, while the other agents share the remaining social power equally; for $t = 1/2$, agent 1 has social power ≈ 0.286 and every other agent ≈ 0.143. Moreover, the computation gives the same values on the two networks. In other words, on these two networks

the self-weight of agent 1 changes how the opinions converge, which depends on the network (Figure 2), but not where they converge.

The bottom panel of Figure 3 shows where the group converges: by point (ii) of Theorem 2.1, the consensus value is $w^\top x(0)$, here computed for $x(0) = (0, 0.2, 0.4, 0.6, 0.8, 1)^\top$, the initial condition of Figure 2. For small t all the agents have almost the same social power and the consensus value is close to the average of the initial opinions, 0.5; as t grows, the social power of agent 1 grows and the consensus value moves towards its initial opinion, $x_1(0) = 0$, which is reached for $t = 1$. For $t = 1/2$ the consensus value is ≈ 0.429, the value reached by the trajectories of Figure 2. Also this curve is the same on the two networks.

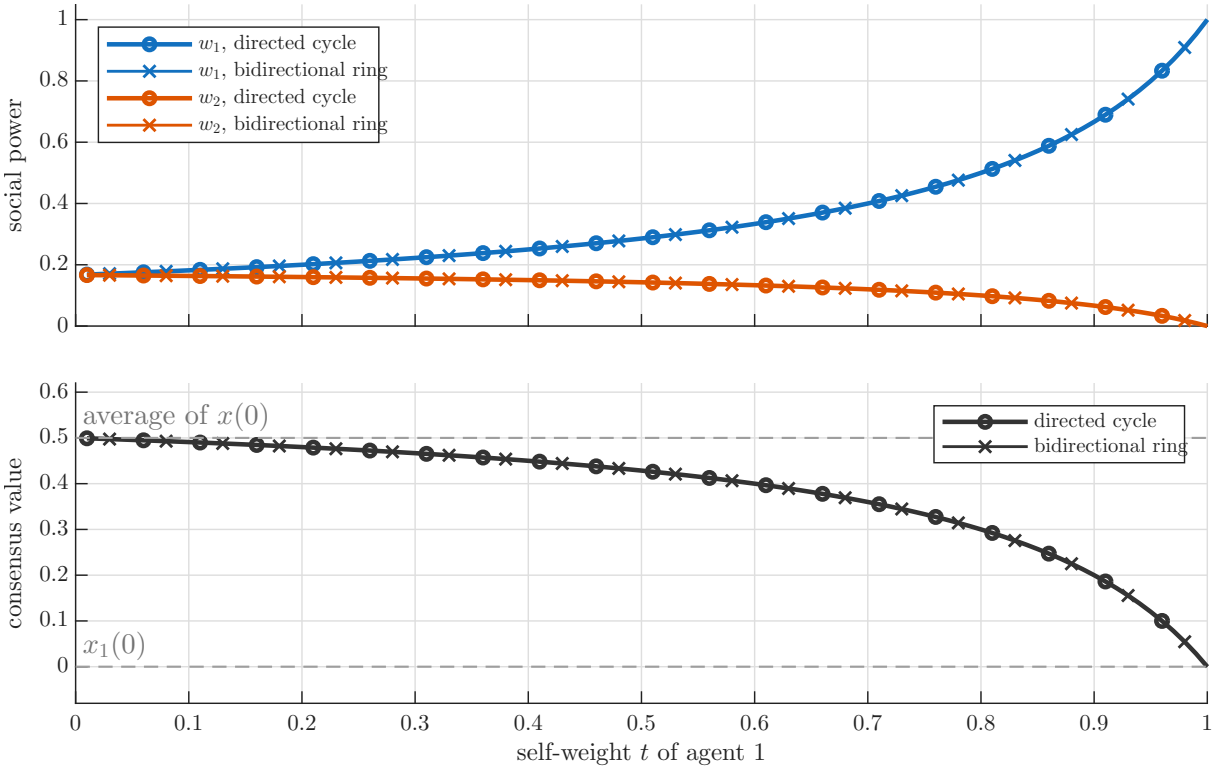


Figure 3: Top: social power of agent 1 (w_1) and of agent 2 (w_2) as a function of the self-weight $t \in (0, 1]$ of agent 1. Bottom: consensus value $w^\top x(0)$ for $x(0) = (0, 0.2, 0.4, 0.6, 0.8, 1)^\top$; the dashed lines mark the average of $x(0)$ and $x_1(0)$. Both panels are computed numerically in the directed cycle (dots) and in the bidirectional ring (crosses); the two networks give the same values.

Remark 2.5 (More than one stubborn agent). If two or more agents are stubborn, each of them is a sink of $C(G)$ made of a single, aperiodic node. If every other agent is influenced, directly or through intermediaries, by at least one stubborn agent, these are the only sinks, and by Theorem 2.3 the group converges, but to a disagreement: every stubborn agent keeps its own initial opinion, and every other agent converges to a convex combination of the opinions of the stubborn agents it is influenced by (see also [5]). Since there is no single consensus value, the social power is no longer defined. This situation, in which several agents are anchored to different fixed opinions, is the starting point of the Friedkin–Johnsen model of Section 3.

3 The Friedkin–Johnsen model

3.1 Introduction to the FJ model

3.1.1 The FJ model

In the FDG model every agent keeps averaging the opinions it listens to, and a persistent disagreement can only come from the structure of the network, i.e. from several closed groups (Theorem 2.3). The Friedkin–Johnsen (FJ) model adds a second ingredient: every agent is also tied to a fixed opinion of its own, called its *prejudice*. Following [5], the model reads

$$x(k+1) = \Lambda W x(k) + (I - \Lambda)\, u, \tag{3}$$

where

- $W \in \mathbb{R}^{n\times n}$ is a row-stochastic matrix, the *matrix of social influences*; with the convention of Section 1.1, $w_{ij} > 0$ means that agent i is influenced by agent j, so W plays the role that A plays in the FDG model;
- $\Lambda = \operatorname{diag}(\lambda_1, \dots, \lambda_n)$, with $\lambda_i \in [0, 1]$ the *susceptibility* of agent i to social influence;
- $u \in \mathbb{R}^n$ is a constant vector collecting the agents' *prejudices*.

Componentwise,

$$x_i(k+1) = \lambda_i \sum_{j=1}^{n} w_{ij}\, x_j(k) + (1 - \lambda_i)\, u_i. \tag{4}$$

At every step agent i averages the opinions it listens to, exactly as in the FDG model, and mixes this average with its prejudice: the weight λ_i goes to the others and the complement $1 - \lambda_i$ goes to u_i. The two quantities play different roles: the prejudice u_i is an opinion and says *where* the agent is anchored, while $1 - \lambda_i$ is a weight and says *how strongly*. The same model appears in [4], where λ_i is called the openness level of agent i and $1 - \lambda_i$ its attachment to the prejudice.

The special cases of (3) are the following [5].

- If $\lambda_i = 1$, agent i follows the FDG rule. If $\Lambda = I$, the FJ model reduces to the FDG model with $A = W$; for this reason we assume $\Lambda \neq I$ from now on.
- If $\lambda_i < 1$, agent i is called *prejudiced*: it is anchored to its prejudice u_i.
- If $\lambda_i = 0$, then $x_i(k) = u_i$ for all $k \geq 1$: the agent ignores the others and jumps to its prejudice after one step. Its opinion never changes, i.e. the agent is stubborn in the sense of Definition 2.4, only if in addition $u_i = x_i(0)$. A stubborn agent of the FJ model is therefore an agent with $\lambda_i = 0$ and $u_i = x_i(0)$.

In social terms, the self-weight of the FDG model and the prejudice of the FJ model are two different forms of resistance to influence. A self-weight ties an agent to its *current* opinion, which still moves towards the others (Section 2.2.1); a prejudice ties it to a *fixed* opinion, which never moves.

In the original formulation of Friedkin and Johnsen the prejudices coincide with the initial opinions, $u = x(0)$, and the susceptibilities are coupled to the self-weights through $1 - \lambda_i = w_{ii}$ [5, 4]. As in [5], we do not impose these conditions: u and $x(0)$ are independent, and so are Λ and W. The choice $u = x(0)$ will be needed in Section 3.1.4 to define the influence centrality.

3.1.2 Convergence and stability

The FJ model (3) is a linear system with the constant input u. Following [5], we say that it *converges* if the limit $x(\infty) = \lim_{k\to+\infty} x(k)$ exists for every initial condition $x(0)$, and that it is *asymptotically stable* if ΛW is convergent, i.e. $\rho(\Lambda W) < 1$ (Definition 1.7 and Theorem 1.8). Following the same reference, a matrix A is called *regular* if $\lim_{k\to+\infty} A^k$ exists, i.e. if it is semi-convergent in the sense of Definition 1.7.

Both properties are characterized by distinguishing the agents that are reached by the prejudices from those that are not.

Definition 3.1 (P-dependent agents [5])**.** An agent is *P-dependent* if it is prejudiced, or if it is influenced, directly or through intermediaries, by some prejudiced agent, i.e. if there is a directed walk in G from the agent to a prejudiced agent. Otherwise, the agent is *P-independent*.

The walk goes from the agent to the prejudiced agent because of our convention on the edges; in [5], where the edges are drawn in the opposite direction, the same walk goes from the prejudiced agent to the agent. In social terms, an agent is P-dependent if at least one prejudice reaches it.

A P-independent agent i is not prejudiced, so $\lambda_i = 1$, and it is not influenced by any P-dependent agent: if $w_{ij} > 0$ with j P-dependent, there would be a walk from i to a prejudiced agent through j, and i would be P-dependent. Since $\Lambda \neq I$, there is at least one prejudiced agent, hence at least one P-dependent agent. Renumbering the agents so that agents $1, \dots, r$, with $r \geq 1$, are P-dependent and agents $r+1, \dots, n$ are P-independent, and splitting $x = [x^1; x^2]$, $u = [u^1; u^2]$, W and Λ accordingly, we have $W^{21} = 0$ and $\Lambda^{22} = I$, and the model (3) decomposes as [5]

$$x^1(k+1) = \Lambda^{11}\left[W^{11}x^1(k) + W^{12}x^2(k)\right] + (I - \Lambda^{11})\,u^1, \tag{5}$$

$$x^2(k+1) = W^{22}x^2(k), \tag{6}$$

where W^{22} is row-stochastic, since W is row-stochastic and $W^{21} = 0$. The P-independent agents therefore follow an FDG model among themselves, with $A = W^{22}$, and no prejudice reaches them; the P-dependent agents receive the opinions of the P-independent ones as an additional input.

Theorem 3.2 ([5])**.** *Let the group have $r \geq 1$ P-dependent agents and $n - r \geq 0$ P-independent ones. Then:*

(i) the subsystem (5) *is asymptotically stable, i.e. $\rho(\Lambda^{11}W^{11}) < 1$;*

(ii) the model (3) *converges if and only if $r = n$ or W^{22} is regular;*

(iii) in this case

$$x^1(\infty) = V\begin{bmatrix} u^1 \\ x^2(\infty) \end{bmatrix}, \qquad V = (I - \Lambda^{11}W^{11})^{-1}\begin{bmatrix} I - \Lambda^{11} & \Lambda^{11}W^{12} \end{bmatrix}, \tag{7}$$

and the matrix V is row-stochastic.

Since V is row-stochastic, the final opinion of every P-dependent agent is a convex combination of the prejudices and of the final opinions of the P-independent agents [5]. The latter are the limit of the FDG model (6): by the Rosetta stone (Section 2.1.4), W^{22} is regular if and only if every sink of the condensation digraph of the subgraph of P-independent agents, whose adjacency matrix is W^{22}, is aperiodic. In social terms, the P-dependent agents settle on a compromise between the prejudices and the opinions of the P-independent agents they listen to whenever these opinions settle; whether the whole group settles is decided only by the P-independent agents.

When $r = n$ there are no P-independent agents and the subsystem (5) is the whole model. This case is the subject of the following result.

Corollary 3.3 ([5])**.** *The FJ model* (3) *is asymptotically stable if and only if all agents are P-dependent. Then, for every* $x(0)$*,*

$$x(\infty) = Vu, \qquad V = (I - \Lambda W)^{-1}(I - \Lambda). \tag{8}$$

This holds, in particular, if $\lambda_i < 1$ *for every agent, or if* $\Lambda \neq I$ *and* G *is strongly connected (*W *irreducible).*

The condition of Corollary 3.3 can also be read on the matrix ΛW with the tools of the course. The matrix ΛW is no longer row-stochastic: since W is row-stochastic, the i-th row of ΛW sums to λ_i. Rows of agents that are not prejudiced sum to 1, rows of prejudiced agents sum to less than 1, and since $\Lambda \neq I$ at least one such row exists. Hence ΛW is row-substochastic (Definition 1.12) and Lemma 1.13 applies.

For $A = \Lambda W$ the weighted out-degree of agent i is λ_i: the nodes with out-degree 1 are the agents that are not prejudiced, and the nodes with out-degree less than 1 are the prejudiced agents. The condition of Lemma 1.13 therefore reads: every agent that is not prejudiced is influenced, directly or through intermediaries, by a prejudiced agent, i.e. all the agents are P-dependent. This is exactly the condition of Corollary 3.3. Indeed, [5] remarks that Corollary 3.3 can be turned into a criterion of Schur stability for substochastic matrices, and [4] proves the same condition for the FJ model through this lemma.

Two observations of ours give an intuition for Corollary 3.3. First, letting $k \to +\infty$ in (3) shows that the limit $x(\infty) = Vu$ is an equilibrium, $x(\infty) = \Lambda W x(\infty) + (I - \Lambda)u$. Letting $e(k) = x(k) - x(\infty)$ and subtracting this equation from (3) gives

$$e(k+1) = \Lambda W\, e(k), \tag{9}$$

an autonomous linear system in which the prejudice u does not appear: ΛW decides whether the opinions converge, u decides where.

Second, the linear system (9) shows the role of the P-dependent agents. At each step, agent i keeps a fraction λ_i of the weighted average of the deviations of the agents it listens to. An agent that is not prejudiced ($\lambda_i = 1$) only averages the deviations, while a prejudiced agent loses a fraction $1 - \lambda_i$ of them (in (3), this fraction is replaced by the agent's fixed prejudice). The deviations therefore leave the group only through the prejudiced agents, and they vanish only if every agent is influenced, directly or through intermediaries, by one of them, i.e. all agents are P-dependent. A group of agents that is not influenced, even indirectly, by any prejudiced agent loses no deviation and behaves as an FDG model, as in (6).

In social terms, the group converges, for every initial opinion, as soon as every agent that is not anchored to a prejudice listens, directly or through intermediaries, to someone who is. On a strongly connected network, such as the two networks of Figure 1, a single prejudiced agent is enough. Note that this does not require aperiodicity, unlike the FDG model of Section 2.2.

If some agents are P-independent, the model is not asymptotically stable, but it may still converge. Part (ii) of Theorem 3.2 gives the condition in terms of W^{22}; in terms of the whole matrix ΛW it reads as follows.

Corollary 3.4 ([5])**.** *The FJ model* (3) *converges if and only if* ΛW *is regular.*

As remarked in [5], this property is not trivial: a system $x(k+1) = Ax(k) + Bu$ with a regular matrix A may not converge, and may even have unbounded solutions, as for $A = B = I$.

3.1.3 Final opinions: consensus or disagreement

When all agents are P-dependent, the final opinions are given by (8), and the convex combination of Theorem 3.2 involves only the prejudices:

$$x_i(\infty) = \sum_{j=1}^{n} v_{ij}\, u_j.$$

Three facts follow.

- The initial opinions are forgotten: $x(0)$ does not appear in $x(\infty)$.
- Only the prejudices of prejudiced agents count: if $\lambda_j = 1$, the j-th column of $I - \Lambda$ is zero, hence so is the j-th column of V, and u_j has no weight in any final opinion.
- Every final opinion lies between the smallest and the largest prejudice of the prejudiced agents.

In the FDG model, the consensus value $w^\top x(0)$ is a weighted average of the initial opinions, with the same weights w for every agent (Theorem 2.1). In the FJ model, the final opinions are weighted averages of the prejudices, and each agent has its own weights, given by its row of V.

3.1.4 Influence centrality

3.2 Stubbornness and social power in the FJ model

3.2.1 Modelling stubbornness: prejudices and stubborn agents

3.2.2 Prejudices, stubbornness and convergence

3.2.3 Prejudices, stubbornness and social power

4 Conclusions

References

[1] Giacomo Baggio. DT averaging systems. Lecture slides, Network Dynamical Systems, University of Padova, 2025.

[2] Giacomo Baggio. Elements of Graph Theory. Lecture slides, Network Dynamical Systems, University of Padova, 2025.

[3] Giacomo Baggio. Elements of Matrix Theory. Lecture slides, Network Dynamical Systems, University of Padova, 2025.

[4] Francesco Bullo. *Lectures on Network Systems*. Kindle Direct Publishing, 1.7 edition, 2024. With contributions by J. Cortés, F. Dörfler, and S. Martínez.

[5] Anton V. Proskurnikov and Roberto Tempo. A tutorial on modeling and analysis of dynamic social networks. Part I. *Annual Reviews in Control*, 43:65–79, 2017.